

SmartVax Cold Chain Guard

Design and simulation of an IoT based vaccine cold storage monitoring and control system using Cisco Packet Tracer

Prepared and submitted by **Mehedi Hasan Mithun**

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INSTRUCTOR	SIMULATION TOOL	PROJECT FILE
Md. Anowar Kabir, Lecturer	Cisco Packet Tracer 9.0	SmartVax_ColdChainGuard.pkt
DATE OF SUBMISSION	IOT DEVICES DEPLOYED	
15 August 2026	9 (minimum required: 4)	

1 Title of the Project

SmartVax Cold Chain Guard: an IoT based vaccine cold storage monitoring, automation and alert system.

The system continuously monitors the temperature, humidity, door state and fire risk of a pharmaceutical cold room, automatically drives the cooling and ventilation equipment to keep the room inside the safe vaccine storage band of 2 to 8 degrees Celsius, and raises an audible and visual alarm the moment the room drifts outside that band or is physically breached. Both a local pharmacist workstation and a remote supervisor on a different subnet can view the live state of every device.

2 Objective

The objectives of this project are as follows.

1. To design a complete IoT network in Cisco Packet Tracer that combines wired infrastructure, a wireless access layer, an IoT registration server and nine IoT end devices.
2. To continuously sense the physical condition of a vaccine cold room using temperature, humidity, smoke and door sensors.
3. To automate the response of the cooling, ventilation, lighting, surveillance and alarm actuators through server side condition rules, with no human intervention required.
4. To hold the storage temperature inside the World Health Organization recommended band of 2 to 8 degrees Celsius using a two threshold control strategy that avoids both spoilage by heat and damage by freezing.
5. To provide immediate alerting when a temperature excursion, a humidity excursion, a fire condition or an unauthorised door opening occurs.
6. To demonstrate secure and addressable network design using WPA2-PSK on the wireless segment, DHCP for the IoT devices, static addressing for infrastructure, and inter subnet routing so that a remote supervisor can monitor the facility.
7. To verify the correctness of the design by simulation, including packet level inspection of the traffic between the sensors and the IoT server.

3 Problem Statement

Vaccines, insulin, blood products and many biological medicines are thermolabile. They remain potent only while they are stored between 2 and 8 degrees Celsius. If the temperature rises above that band the active agent degrades, and if it falls below freezing the vaccine is destroyed outright. Neither type of damage is visible: a spoiled vial looks exactly like a good one. The failure is therefore silent, and it is discovered only when patients fail to develop immunity.

This chain of custody, from manufacturer to refrigerator to patient, is known as the cold chain. The World Health Organization has repeatedly identified cold chain failure as one of the largest sources of vaccine wastage worldwide, and the losses are concentrated at the last storage step, that is, the district hospital or rural clinic

refrigerator. The practical causes are mundane:

- A power cut at night, with nobody present to notice that the compressor has stopped.
- A cold room door left ajar after a stock movement, so warm outside air enters for hours.
- A thermostat that fails and overcools the room, freezing the entire stock.
- High humidity that causes condensation, corroding the equipment and damaging vial labels.
- Manual temperature logging done twice a day on a paper chart, which cannot detect an excursion that begins and ends between two readings.
- No alerting mechanism at all, so even a detected problem is not escalated until the next working morning.

The core problem is that traditional cold storage is monitored by people, on a schedule, and people are absent for most of the day. What is required is a system that senses continuously, decides automatically and alerts immediately.

SmartVax Cold Chain Guard addresses exactly this gap. It replaces periodic manual observation with a permanently connected sensor network, replaces human judgement with deterministic automation rules held on an IoT server, and replaces the delayed paper escalation path with an immediate local alarm plus a remotely visible dashboard.

4 Components Used

4.1 IoT end devices

#	DEVICE	LABEL	TYPE	ROLE IN THE SYSTEM
1	Temperature Monitor	TEMP-01	Sensor	Reads the cold room air temperature in degrees Celsius. Primary input to the cooling and alarm logic.
2	Humidity Monitor	HUM-01	Sensor	Reads relative humidity. Drives the ventilation logic and protects against condensation.
3	Smoke Detector	SMOKE-01	Sensor	Detects smoke or fire in the storage area.
4	Door	Door-01	Sensor and actuator	Reports the open or closed state of the cold room door. Acts as the security and excursion trigger.
5	Air Conditioner	AC-01	Actuator	The primary cooling unit. Switched on and off by the temperature band rules.
6	Ceiling Fan	FAN-01	Actuator	Ventilation. Runs at High to remove moist air when humidity climbs.
7	Siren	SIREN-01	Actuator	Audible alarm for heat excursion, freezing, fire and door ajar events.
8	Webcam	CAM-01	Actuator	Surveillance. Activates automatically whenever the door is opened, so every access is recorded.

#	DEVICE	LABEL	TYPE	ROLE IN THE SYSTEM
9	Light	LIGHT-01	Actuator	Room lighting. Follows the door state, giving safe access and saving energy.

Table 1. Nine IoT end devices. The assignment requires a minimum of four.

4.2 Networking and computing components

COMPONENT	MODEL IN PACKET TRACER	LABEL	CONFIGURATION
Router	Cisco ISR 2911	CC-Router	G0/0 192.168.10.1/24 G0/1 192.168.20.1/24
Access layer switch	Cisco Catalyst 2960	CC-SW1	Cold room and server segment
Access layer switch	Cisco Catalyst 2960	CC-SW2	Supervisor office segment
Wireless access point	AccessPoint-PT	ColdRoom-AP	SSID SmartVax_IoT, WPA2-PSK with AES
IoT registration server	Server-PT	IoT-Server	192.168.10.2/24 IoT, DHCP, DNS and HTTP services
Local workstation	PC-PT	Pharmacist-PC	192.168.10.3/24 static
Remote workstation	Laptop-PT	Supervisor-Laptop	192.168.20.10/24 static
Cabling	Copper straight through	not applicable	Router to switches, switches to server, PC, laptop and access point
Wireless links	IEEE 802.11	not applicable	All nine IoT devices, each fitted with a PT-IOT-NM-1W adapter

Table 2. Network infrastructure and computing components.

4.3 Addressing plan

SEGMENT	NETWORK	GATEWAY	ASSIGNMENT METHOD	HOSTS
Cold Room LAN	192.168.10.0/24	192.168.10.1	Static for infrastructure, DHCP from 192.168.10.100 for IoT devices	Server, pharmacist workstation, access point, 9 IoT devices
Admin LAN	192.168.20.0/24	192.168.20.1	Static	Supervisor laptop

Table 3. IP addressing plan. The two segments are joined by directly connected routing on CC-Router.

5 System Architecture and Network Diagram

The system is built in three tiers, which is the standard IoT reference model applied to a single facility.

- **Perception tier.** The nine IoT end devices. Four of them sense the physical environment and five of them act on it.
- **Network tier.** The wireless access point carries all IoT traffic onto the wired LAN. Two switches provide the access layer and the router joins the two subnets and provides the default gateway for each.
- **Application tier.** The IoT server holds the device registry, the automation rule engine and the web dashboard. Client browsers on the pharmacist workstation and on the remote supervisor laptop consume that dashboard.

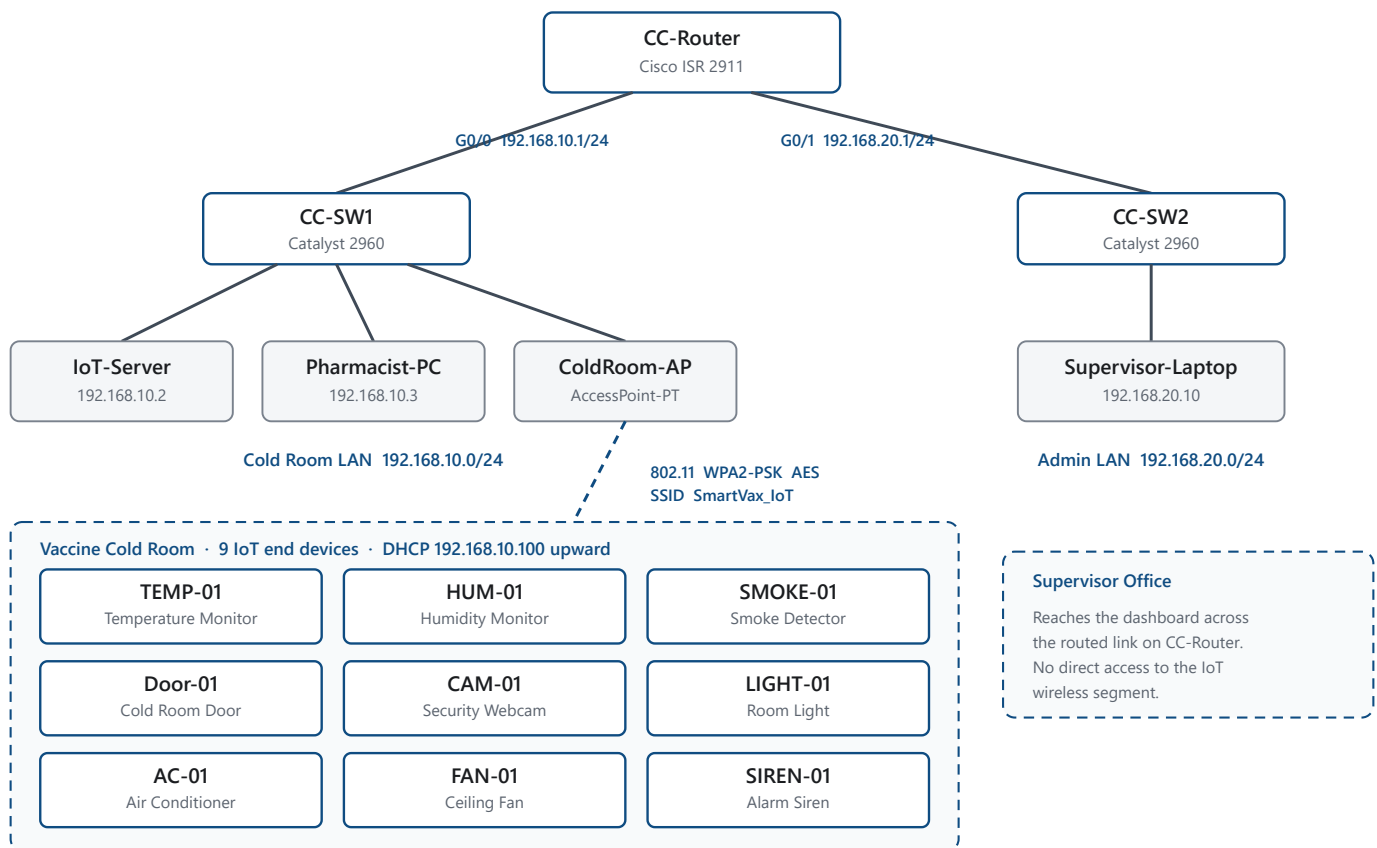


Figure 1. Logical topology of SmartVax Cold Chain Guard. Solid lines are copper straight through links, dashed blue is the 802.11 wireless association. All nine IoT devices associate to ColdRoom-AP, receive an address by DHCP from IoT-Server, and register to the IoT registration service on 192.168.10.2.

5.1 Why this topology

Three design decisions are worth justifying.

A registration server rather than a home gateway. Packet Tracer allows IoT things to be attached directly to a home gateway, but that confines the whole project to a single flat segment and hides the network engineering. Using a Server-PT with the IoT service enabled means the devices must obtain addressing, resolve and reach a server, authenticate with a user account and register, which exercises DHCP, HTTP and the registration protocol, and it places the rule engine on a device that is architecturally separate from the sensors.

An access point rather than a wireless router. The access point is a layer two bridge, so every IoT device sits in the same subnet as the server. A wireless router would introduce network address translation between the sensors and the server for no benefit, and would make the remote monitoring path harder to explain.

A second routed subnet. Placing the supervisor on `192.168.20.0/24` demonstrates that the dashboard is reachable from outside the cold room segment, which is what a real deployment needs, while still keeping the IoT wireless network isolated behind the router.

6 Working Principle

The system runs a continuous sense, decide, act cycle. There is no polling by a human operator at any point.

- 1 Association and addressing.** On power up, each IoT device associates with `ColdRoom-AP` using WPA2-PSK, then broadcasts a DHCP discover. `IoT-Server` answers from the pool starting at `192.168.10.100` and supplies the gateway `192.168.10.1` and the DNS server.
- 2 Registration.** Each device is configured with a remote IoT server address of `192.168.10.2` and the credentials of the `admin` account. It registers itself, publishes the list of properties it exposes, such as temperature or open state, and from that moment appears as a card on the server dashboard.
- 3 Telemetry.** Sensors read the simulated environment continuously and push their current value to the server. The server keeps the latest value for every property of every registered device.
- 4 Rule evaluation.** On every update the server evaluates the twelve conditions in the rule table. A condition is a boolean expression over one or more device properties. This is the decision layer, and it lives on the server rather than in the devices, so a rule can be added or a threshold changed without touching any device.
- 5 Actuation.** When a condition evaluates true the server sends a set command to the target actuator. The air conditioner, fan, light, webcam or siren changes state, and the change is visible immediately both on the topology canvas and on the dashboard.
- 6 Feedback.** Because the air conditioner and the fan physically change the environment that the sensors read, the system forms a closed control loop. The temperature falls, the sensor reports the lower value, the rule that turned the cooling on stops matching and the reset rule turns it off.
- 7 Monitoring and escalation.** The pharmacist sees the live state locally. The supervisor sees exactly the same dashboard across the routed link from `192.168.20.0/24` . The siren provides on site escalation for anyone in the building.

6.1 The control strategy

The temperature logic is deliberately built as a band with hysteresis rather than a single setpoint, and this is the most important engineering detail in the project.

TEMPERATURE	ZONE	SYSTEM RESPONSE
below 2 °C	Freezing, vaccine destroyed	Cooling off, siren on

TEMPERATURE	ZONE	SYSTEM RESPONSE
2 to 6 °C	Safe, cooling not needed	Cooling off, siren off
6 to 8 °C	Safe operating band	Cooling holds its current state
8 to 10 °C	Warm, corrective action	Cooling on, no alarm yet
above 10 °C	Excursion, stock at risk	Cooling on, siren on

Table 4. Temperature control bands as finally tuned. The gap between the cool on threshold of 8 °C and the cool off threshold of 6 °C is the hysteresis. Section 9.2 explains why the cut off was raised from an initial 4 °C.

Without that gap the air conditioner would switch on and off many times per minute around a single threshold, which in a real installation destroys the compressor. Separating the alarm threshold of 10 degrees from the cooling threshold of 8 degrees means the system is given a chance to correct the problem quietly before it disturbs anybody, and the siren then indicates a genuine failure rather than normal operation.

7 Automation Rules

Twelve rules were implemented on the IoT server. The assignment requires a minimum of one.

#	RULE NAME	CONDITION	ACTION	PURPOSE
1	R1-Cooling-ON	TEMP-01 Temperature > 8.0 °C	Set AC-01 On to true	Start cooling before the safe band is exceeded
2	R2-Cooling-OFF	TEMP-01 Temperature < 6.0 °C	Set AC-01 On to false	Stop cooling with hysteresis, prevents freezing and compressor cycling
3	R3-Heat-Alarm	TEMP-01 Temperature > 10.0 °C	Set SIREN-01 On to true	Escalate a genuine heat excursion
4	R4-Freeze-Alarm	TEMP-01 Temperature < 2.0 °C	Set SIREN-01 On to true	Escalate a freezing event, which is equally destructive
5	R5-Alarm-Reset	Match all: TEMP-01 Temperature ≥ 3.0 °C TEMP-01 Temperature ≤ 8.0 °C	Set SIREN-01 On to false	Silence the alarm once the room is safe again

#	RULE NAME	CONDITION	ACTION	PURPOSE
6	R6-Ventilation-ON	HUM-01 Humidity > 60 %	Set FAN-01 Status to High	Extract moist air, prevent condensation on the vials
7	R7-Ventilation-OFF	HUM-01 Humidity < 45 %	Set FAN-01 Status to Off	Stop ventilation once humidity is normal
8	R8-Door-Light	Door-01 Open is true	Set LIGHT-01 Status to On	Safe access without a manual switch
9	R9-Door-Camera	Door-01 Open is true	Set CAM-01 On to true	Record every physical access to the stock
10	R10-Door-Light-OFF	Door-01 Open is false	Set LIGHT-01 Status to Off	Energy saving and reduced heat load
11	R11-Door-Ajar-Excursion	Match all: Door-01 Open is true TEMP-01 Temperature > 8.0 °C	Set SIREN-01 On to true	Composite two sensor rule. Catches the most common real cause of spoilage, a door left open
12	R12-Fire-Alarm	SMOKE-01 Level > 0.5	Set SIREN-01 On to true	Life safety and asset protection

Table 5. The complete automation rule set as configured in the Conditions editor of the IoT server.

Note on rule design. Rules 2, 5, 7 and 10 are reset rules. Every rule that switches an actuator on is paired with a rule that switches it off again under a clearly defined and non overlapping condition. Without such pairs an actuator latches into the on state permanently after the first trigger, and the system can never return to normal. Rule 11 is a composite rule that combines a binary sensor with an analogue one, and it detects a failure that neither sensor could detect alone: a door left open is only a problem once it has actually started to warm the room.

8 Screenshots of the Simulation

The figures below were captured from the running simulation of `SmartVax_ColdChainGuard.pkt` in Cisco Packet Tracer 9.0. Together they evidence the build, the registration of all nine IoT devices, the full rule set, and each automated behaviour described in section 9.

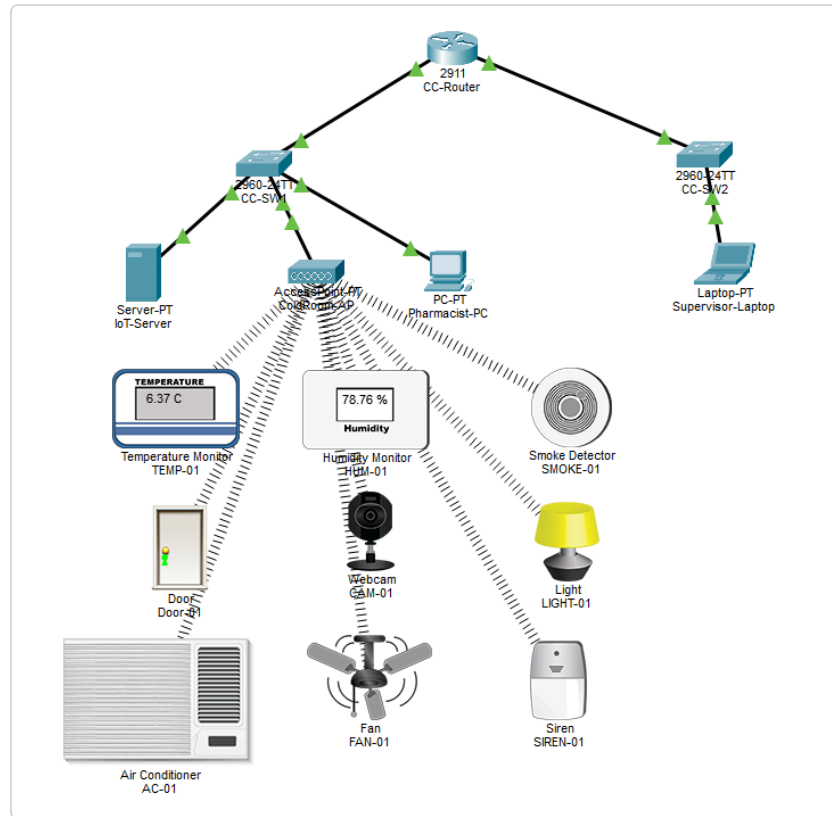


Figure 2. Logical topology as built. The routed core of CC-Router, CC-SW1 and CC-SW2 sits above the wired hosts, and the nine IoT devices below connect by dashed 802.11 links to ColdRoom-AP. At the moment of capture the room was at 6.37 degrees Celsius with humidity at 78.76 percent and the ceiling fan running.

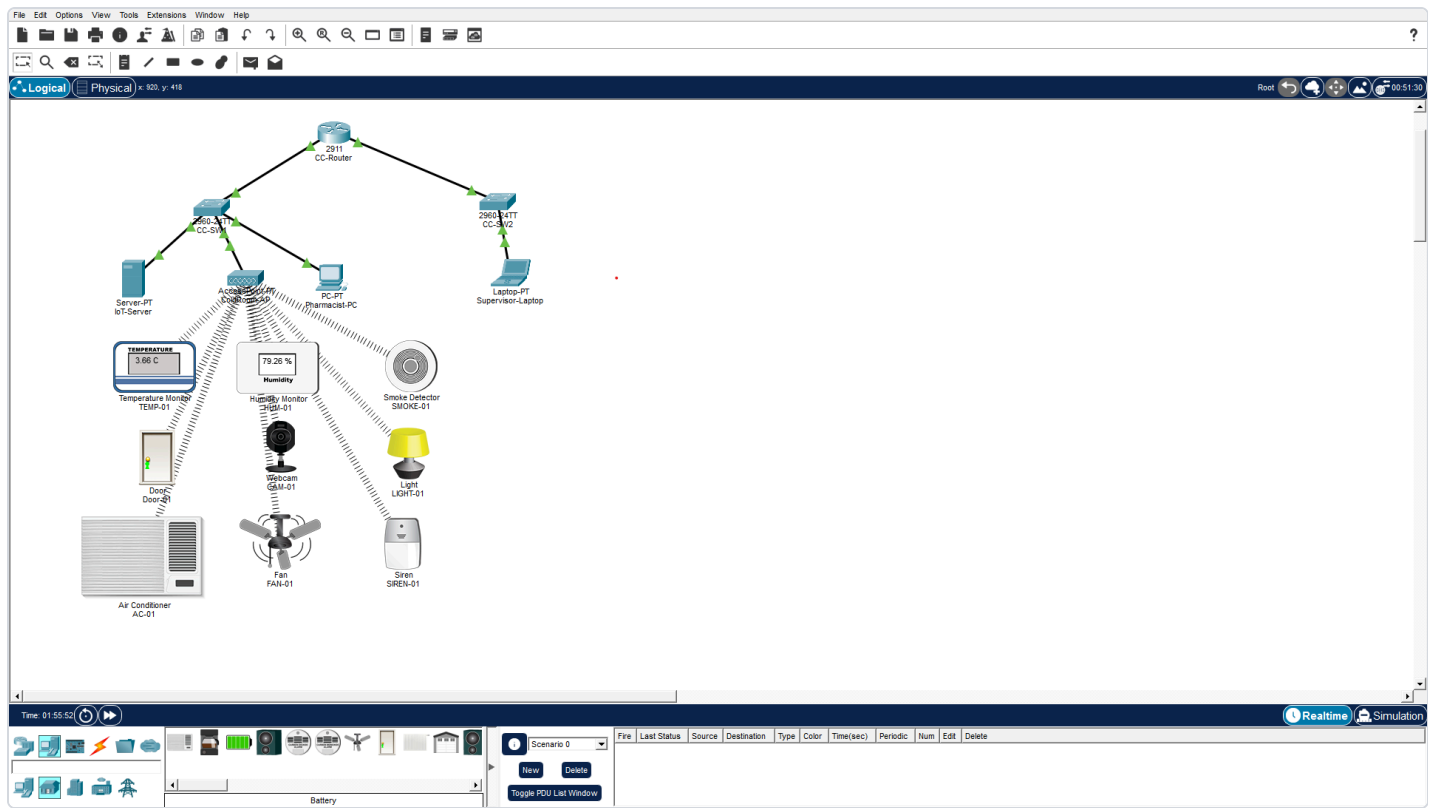


Figure 3. The same topology in the full Packet Tracer workspace, showing the design in Logical view under Realtime mode with the room at 3.66 degrees Celsius.

```

CC-Router
Physical Config CLI Attributes
IOS Command Line Interface

program load complete, entry point: 0x00000000, size: 0x100000
IOS Image Load Test
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
TYPE MEMORY_REQ
Onboard devices &
buffer pools 0x022f6000
TOTAL: 0x022f6000
Rounded IOMEM up to: 38Mb
Using 6 percent iomem. (380b/6120b)

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Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400K3
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```

Figure 4. CC-Router console. The closing lines confirm the line protocol on both GigabitEthernet0/0 and GigabitEthernet0/1 changing state to up, which brings the 192.168.10.0/24 and 192.168.20.0/24 gateways into service.

IoT-Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: **FastEthernet0** Service: ☒ On ☐ Off

Pool Name: **serverPool**

Default Gateway: **192.168.10.1**

DNS Server: **192.168.10.2**

Start IP Address: **192** **168** **10** **100**

Subnet Mask: **255** **255** **255** **0**

Maximum Number of Users: **50**

TFTP Server: **0.0.0.0**

WLC Address: **0.0.0.0**

Add **Save** **Remove**

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	192.168.10.2	192.168.10.100	255.255.255.0	50	0.0.0.0	0.0.0.0

☐ Top

Figure 5. IoT-Server DHCP service. The serverPool is enabled and issues addresses from 192.168.10.100 with gateway 192.168.10.1 and DNS 192.168.10.2, which is how every IoT device obtains its address.

IoT-Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

Registration Server

This service runs on top of the HTTP or HTTPS service.

Service: ☒ On ☐ Off

	Username	Password
1	admin	admin

Delete

☐ Top

Figure 6. IoT-Server registration service. The service runs on top of HTTP and holds the admin account that all nine devices authenticate against when they register.

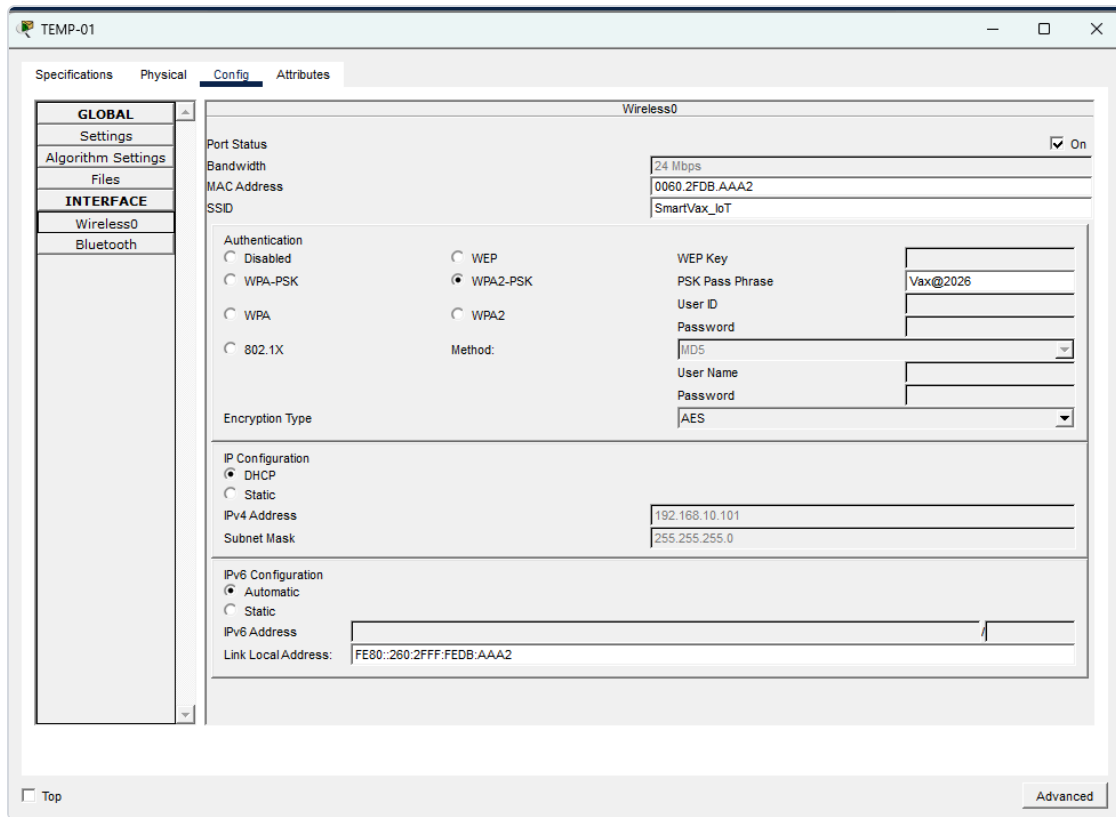


Figure 7. TEMP-01 wireless interface, representative of all nine IoT devices. It is associated with SSID SmartVax_IoT under WPA2-PSK with AES encryption, and has received 192.168.10.101 by DHCP.

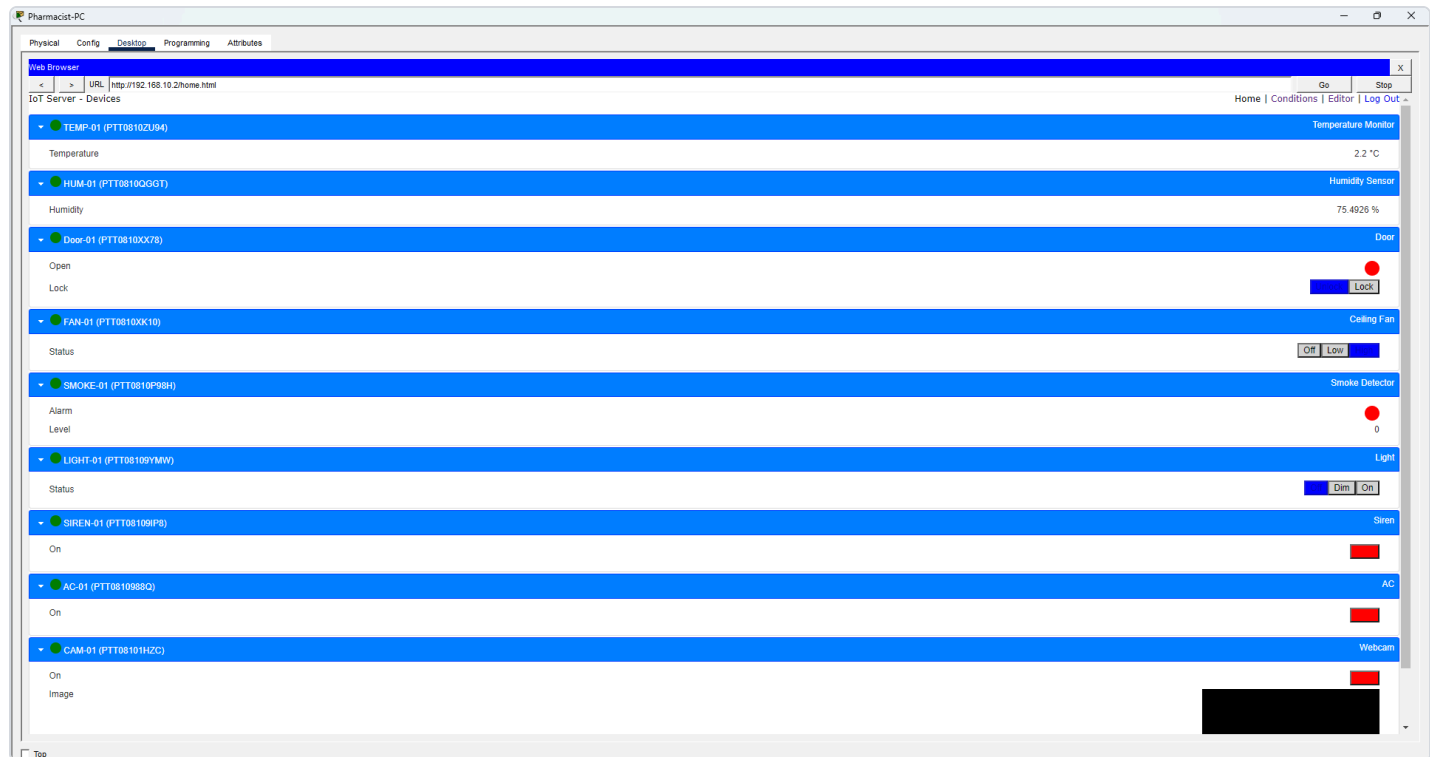


Figure 8. The IoT dashboard on Pharmacist-PC listing all nine registered devices with live values. The room is at 2.2 degrees Celsius with the cooling off and the siren off, near the bottom of the safe band during a cooling coast down but still above the 2 degree freeze threshold, and the ceiling fan is running at High against 75.49 percent humidity.

Pharmacist-PC

Physical Config Desktop Programming Attributes

Web Browser

URL: http://192.168.10.2/conditions.html

Go Stop

Home | Conditions | Editor | Log Out

Actions		Enabled	Name	Condition	Actions
Edit	Remove	Yes	R1-Cooling-ON	TEMP-01 Temperature > 8.0 °C	Set AC-01 On to true
Edit	Remove	Yes	R2-Cooling-OFF	TEMP-01 Temperature < 6.0 °C	Set AC-01 On to false
Edit	Remove	Yes	R3-Heat-Alarm	TEMP-01 Temperature > 10.0 °C	Set SIREN-01 On to true
Edit	Remove	Yes	R8-Door-Light	Door-01 Open is true	Set LIGHT-01 Status to On
Edit	Remove	Yes	R4-Freeze-Alarm	TEMP-01 Temperature < 2.0 °C	Set SIREN-01 On to true
Edit	Remove	Yes	R5-Alarm-Reset	Match all: • TEMP-01 Temperature >= 3.0 °C • TEMP-01 Temperature <= 8.0 °C	Set SIREN-01 On to false
Edit	Remove	Yes	R6-Ventilation-ON	HUM-01 Humidity > 60 %	Set FAN-01 Status to High
Edit	Remove	Yes	R7-Ventilation-OFF	HUM-01 Humidity < 45 %	Set FAN-01 Status to Off
Edit	Remove	Yes	R9-Door-Camera	Door-01 Open is true	Set CAM-01 On to true
Edit	Remove	Yes	R10-Door-Light-OFF	Door-01 Open is false	Set LIGHT-01 Status to Off
Edit	Remove	Yes	R11-Door-Ajar-Excursion	Match all: • Door-01 Open is true • TEMP-01 Temperature > 8.0 °C	Set SIREN-01 On to true
Edit	Remove	Yes	R12-Fire-Alarm	SMOKE-01 Level > 0.5	Set SIREN-01 On to true

Add

Figure 9. The complete rule set on the Conditions page. All twelve rules are enabled, including the two composite Match all rules, R5-Alarm-Reset and R11-Door-Ajar-Excursion.

Pharmacist-PC

Physical Config Desktop Programming Attributes

Web Browser

URL: http://192.168.10.2/home.html

Go Stop

Home | Conditions | Editor | Log Out

IoT Server - Devices

- TEMP-01 (PTT0810ZU94) Temperature Monitor
 - Temperature 7.6 °C
- HUM-01 (PTT0810QGOT) Humidity Sensor
 - Humidity 78.5135 %
- Door-01 (PTT0810XX78) Door
 - Open
 - Lock
- FAN-01 (PTT0810XK10) Ceiling Fan
 - Status Off Low
- CAM-01 (PTT0810HZC) Webcam
- SMOKE-01 (PTT0810P9BH) Smoke Detector
 - Alarm Level 0
- LIGHT-01 (PTT08109YMW) Light
 - Status Dim On
- SIREN-01 (PTT08109P78) Siren
 - On
- AC-01 (PTT081098RQ) AC
 - On

Figure 10. Steady state inside the safe band. The room is at 7.6 degrees Celsius, AC-01 is off and SIREN-01 is off. No rule is calling for action on the temperature loop.

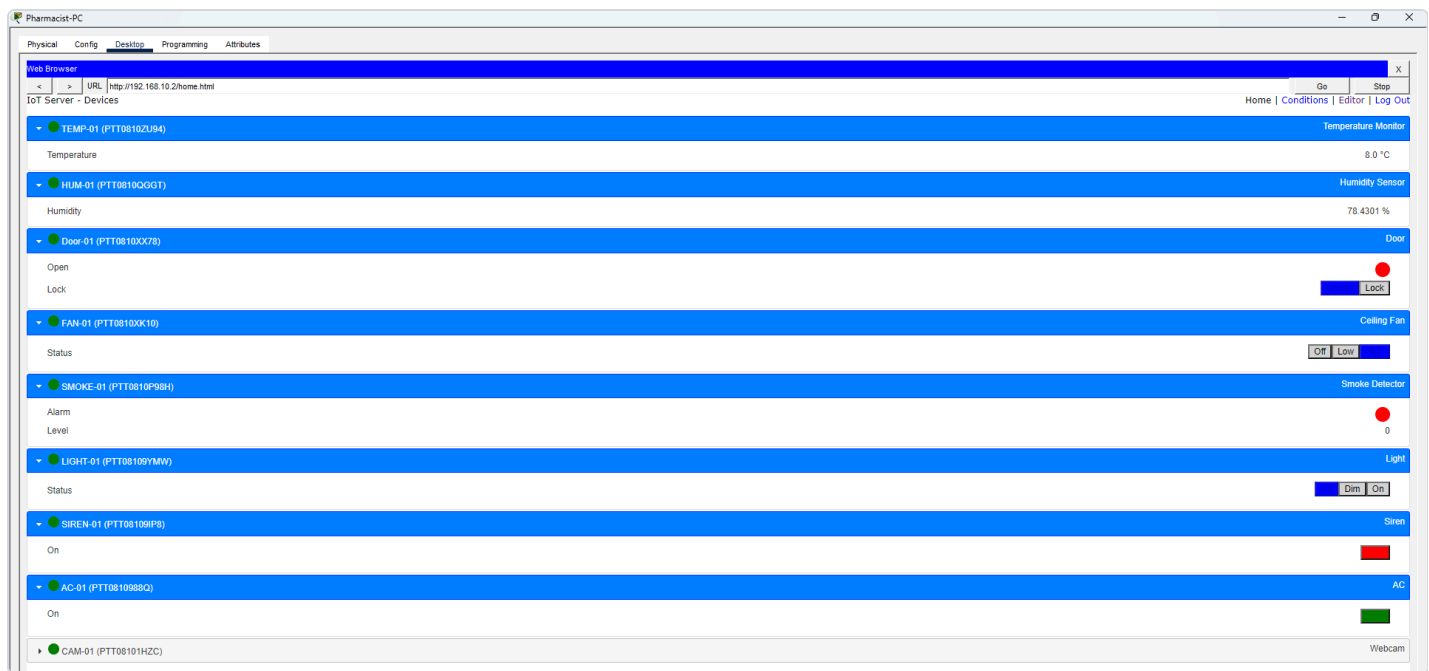


Figure 11. R1-Cooling-ON firing. The room has reached 8.0 degrees Celsius and AC-01 has switched on automatically, shown by the green indicator, with no operator action.

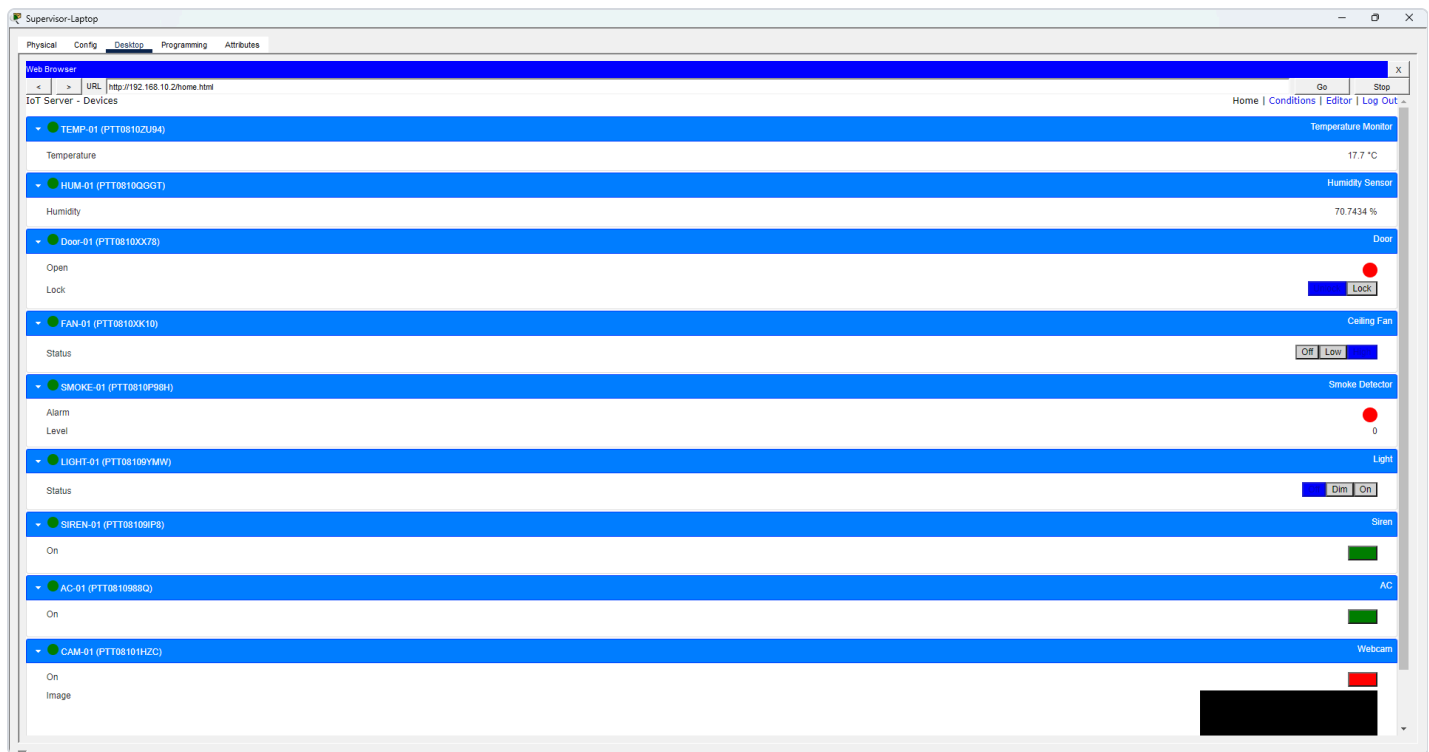


Figure 12. Heat excursion, R3-Heat-Alarm. During the warm up from the raised ambient the room reached 17.7 degrees Celsius, well past the 10 degree threshold. R1 has the air conditioner running and R3 has activated the siren, both shown green. This is the escalation path for a cooling system that cannot keep up.



Figure 13. Door triggered security and lighting, R8 and R9. Opening Door-01 sets Open to true, and within the same evaluation cycle LIGHT-01 has switched on and CAM-01 has switched on. The webcam image confirms the camera is live and has captured the person entering.

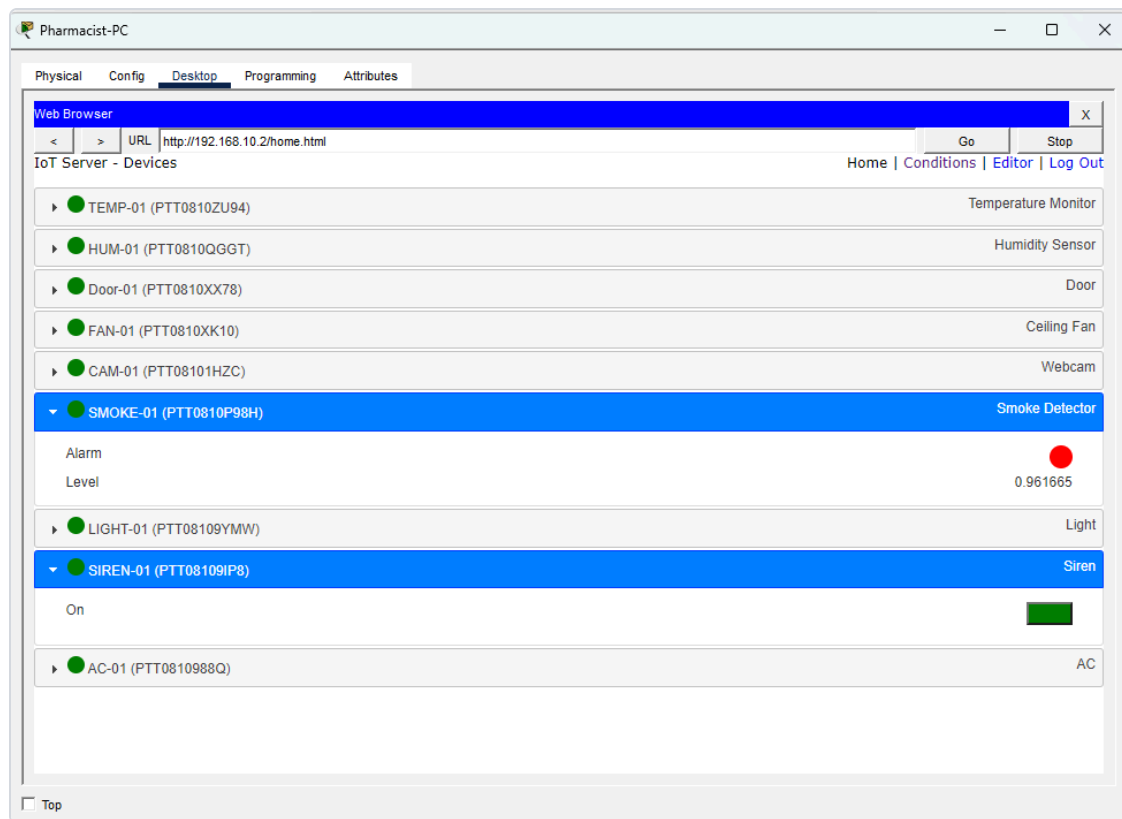


Figure 14. Fire detection, R12-Fire-Alarm. SMOKE-01 reports a smoke level of 0.961665, well past the 0.5 threshold, and SIREN-01 has been switched on automatically by the rule engine.

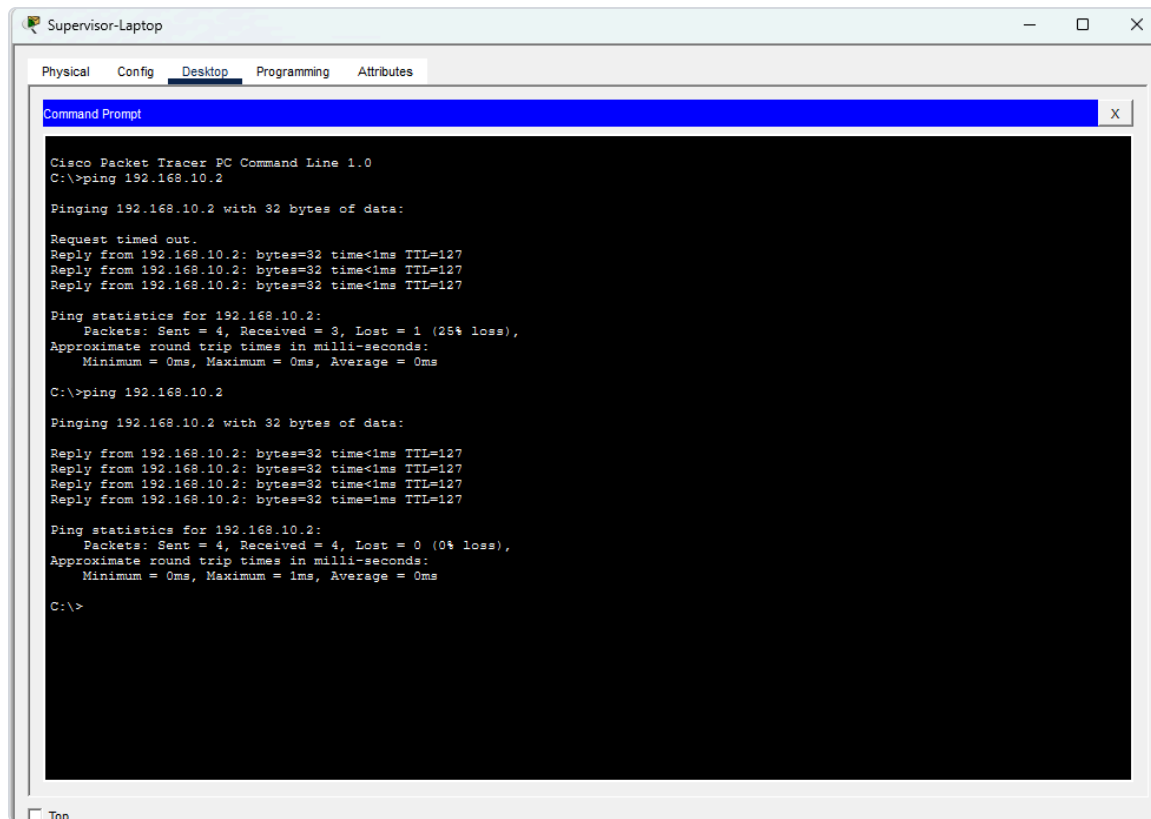


Figure 15. Inter subnet reachability from Supervisor-Laptop at 192.168.20.10. The second attempt returns four replies with no loss at TTL 127, one less than the initial 128, which confirms the traffic crossed exactly one router. The single lost packet on the first attempt is ARP resolution.

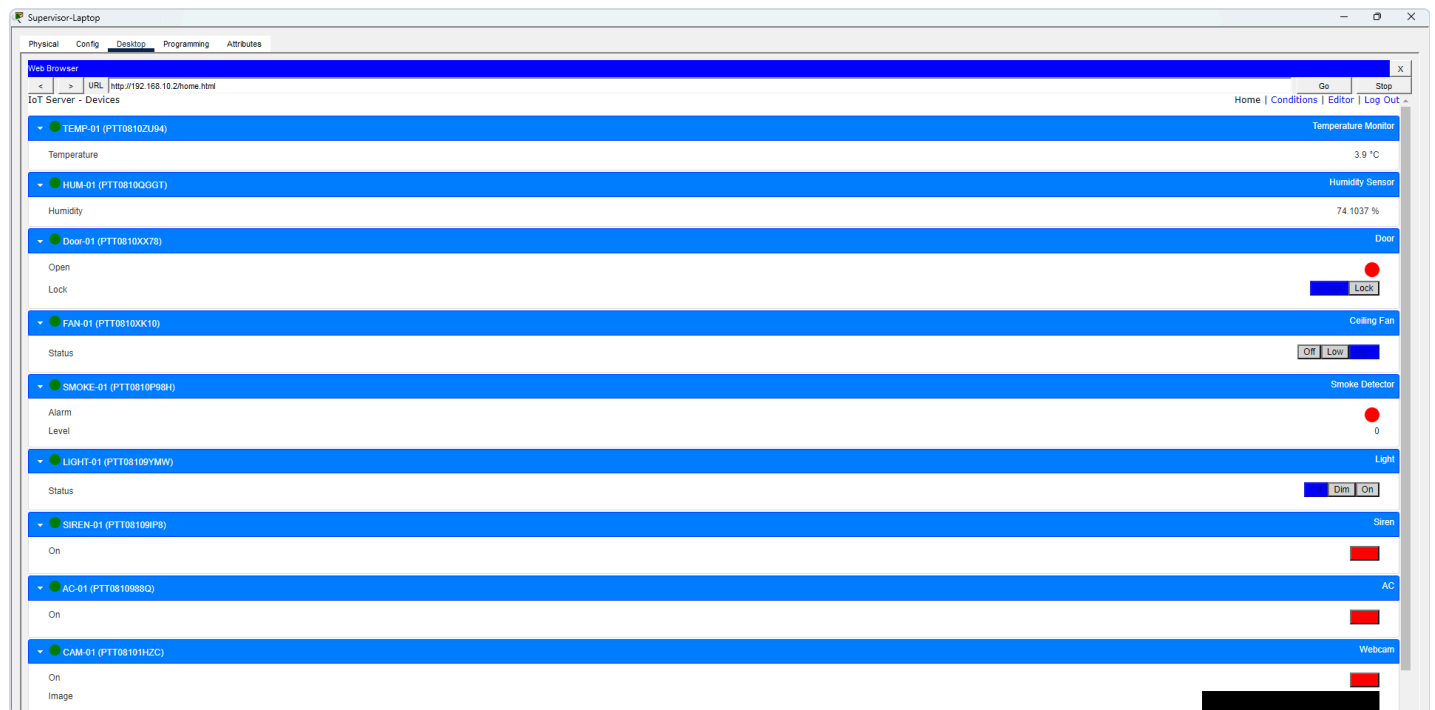


Figure 16. Remote monitoring. The same dashboard rendered in the browser on Supervisor-Laptop from the separate 192.168.20.0/24 subnet, showing the room at 3.9 degrees Celsius. The supervisor sees the full facility without any access to the IoT wireless segment.

9 Results and Discussion

9.1 Test results

TEST	STIMULUS APPLIED	EXPECTED OUTCOME	OBSERVED	RESULT
Wireless association	Nine IoT devices fitted with a PT-IOT-NM-1W adapter and joined to SSID SmartVax_IoT under WPA2-PSK	All associate and receive a DHCP address from 192.168.10.100 upward	All nine associated. AC-01 received 192.168.10.101	Pass
Device registration	Remote Server set to 192.168.10.2 with the admin account on each device	All nine appear on the IoT dashboard	Nine devices listed with live property values	Pass
Inter subnet routing	ping 192.168.10.2 from 192.168.20.10	Four ICMP replies	Four replies at TTL 127, confirming exactly one router hop. The first attempt lost one packet to ARP resolution, as expected	Pass
Cooling control, R1	Ambient temperature set to 20 °C so the room warms past 8 °C	AC-01 switches on	AC-01 switched on at 8 °C and the temperature began falling	Pass
Cooling cut off, R2	Temperature falls back to the cut off threshold	AC-01 switches off	AC-01 switched off at 6 °C	Pass
Hysteresis and band holding	System left free running for several cycles	Stable cycling with no rapid switching, temperature held inside the safe band	Steady oscillation between 6.0 and 8.3 °C . No chattering observed	Pass
Alarm reset, R5	Temperature re-enters the 3 to 8 °C range	SIREN-01 switches off without operator action	Siren silenced automatically on every cycle	Pass
Freeze alarm, R4	Ambient temperature at 0 °C, room allowed to fall	SIREN-01 on, AC-01 off	Room reached -0.03 °C, siren activated and AC-01 remained off	Pass
Heat excursion alarm, R3	Requires the room to exceed 10 °C	SIREN-01 switches on	Observed at 17.7 °C during the warm up from the raised ambient. SIREN-01 activated while AC-01 was running (figure 12)	Pass
Ventilation on, R6	Ambient humidity at approximately 75 percent	FAN-01 switches to High	FAN-01 ran at High continuously	Pass
Ventilation off, R7	Requires humidity below 45 percent	FAN-01 switches off	Not reached. The Packet Tracer Ceiling Fan does not alter simulated humidity, so the ventilation loop cannot close. See section 9.3	Not triggered
Door security, R8, R9 and R10	Alt click on Door-01	LIGHT-01 and CAM-01 switch on, both clear when the door closes	Door-01 Open went true, LIGHT-01 switched On and CAM-01 switched On with a live image (figure 13)	Pass

TEST	STIMULUS APPLIED	EXPECTED OUTCOME	OBSERVED	RESULT
Composite rule, R11	Door opened while the temperature is above 8 °C	SIREN-01 switches on	Configured and enabled, but not exercised. The tuned control loop peaks at 8.3 °C, so the window in which the rule can fire is only 0.3 °C wide and was not captured. Its two constituent behaviours, the door trigger and the temperature comparison, are both confirmed independently	Not exercised
Fire detection, R12	Alt click on SMOKE-01 to raise the smoke level above 0.5	SIREN-01 switches on	SMOKE-01 reported a level of 0.961665 and SIREN-01 switched on immediately (figure 14)	Pass

Table 6. Functional test matrix. Twelve behaviours confirmed in simulation. One rule could not be triggered and one was not exercised, for the reasons recorded in the Observed column.

9.2 Discussion

The closed control loop is the most significant result. Because the air conditioner alters the same simulated environment that the temperature sensor reads, the system does not merely react once, it settles. With ambient held at 20 °C the room warms, R1 starts the cooling at 8 °C, the temperature falls, R2 stops the cooling at 6 °C, the room warms again, and the cycle repeats indefinitely between **6.0 and 8.3 °C** with no operator action at any point. That self sustaining oscillation, rather than any single alarm, is what distinguishes a control system from an alerting system.

Two tuning problems were found during testing, and both were worth finding.

Thermal overshoot. The cut off threshold in R2 was initially set to 4 °C. In simulation the room did not stop at 4, it coasted down to **-0.03 °C** and tripped the freeze alarm. The cooling already present in the air continues to act after the unit is commanded off, so the controller must anticipate rather than react. Raising the cut off from 4 °C to 6 °C absorbed the overshoot and the room has since held the band on every cycle. This is exactly the lag problem that makes simple on off control unsuitable for tightly toleranced storage, and it is the argument for proportional control in a production installation.

No thermal load. With the default environment the ambient temperature sat at 0 °C, so the room drifted to freezing whenever cooling stopped and the cooling rules had nothing to act against. Setting the ambient to 20 °C gave the loop a realistic heat load to overcome. This mirrors reality: a cold room is defined by the difference between the outside temperature and the target, and a system tested without that difference has not really been tested.

The escalation path. R3 raises the siren above 10 °C and was observed firing at **17.7 °C** during the initial warm up from the raised ambient, while the air conditioner was already running. Once the loop reached its steady cycle it stopped firing, because the cooling caps the room at 8.3 °C and the escalation threshold is no longer reached. That transition is the point of the design. Separating the correction threshold of 8 °C from the alarm threshold of 10 °C means the room is given a chance to fix itself quietly, and the siren sounds only when automatic correction has failed to keep up. A system that alarms the instant any threshold is crossed trains its operators to ignore it; one that alarms only on genuine failure carries real information.

Open loop versus closed loop. The temperature subsystem is closed loop, the humidity subsystem is not. R6 correctly switches the ceiling fan to High whenever humidity exceeds 60 percent, and this was observed throughout. However the Packet Tracer Ceiling Fan has no effect on the simulated humidity, so the fan can never reduce the reading that started it, and R7 can never fire. The system detects and acts but cannot observe its own correction. The distinction matters because an open loop actuator gives no evidence that the intervention worked, which in a real facility is the difference between a control system and an indicator lamp.

Device class matters more than device name. The first cooling unit placed in the design was an Air Cooler taken from the Components library. It registered to the server successfully and appeared on the dashboard, but reported *No remote control API* and never appeared in the Conditions editor, because components are intended to be wired to a microcontroller rather than addressed over the network. Replacing it with the Air Conditioner from the smart device library resolved this. The general lesson is that network reachability and registration are necessary but not sufficient for automation; the device must also publish a controllable property.

The composite rule. R11 deserves particular mention, and its one weakness is instructive. Neither an open door nor a warm room is individually conclusive: the door is legitimately opened many times a day, and a brief warm reading follows any access. The conjunction of the two is a strong signal that the door has been left open long enough to matter, and it is precisely the failure mode that causes most real cold chain losses. In testing, however, the rule was never observed to fire, because the tuned control loop is good enough that the room only spends a fraction of a degree above the 8 °C trigger before the cooling pulls it back. A well behaved plant leaves a narrow window for a rule that depends on the plant misbehaving. In a production system R11 would be given a time qualifier instead, firing when the door has been open for some minutes rather than at the instant a threshold is crossed, which is the same limitation discussed in section 9.3.

9.3 Limitations

- The Packet Tracer Ceiling Fan does not alter the simulated humidity, so the ventilation subsystem is open loop and R7 cannot be exercised without editing the environment by hand. A real extractor fan would lower the humidity its own sensor reads.
- Only devices from the smart device library publish a controllable property. Parts taken from the Components library register to the server but report no remote control API and cannot be driven by any rule, which constrains the choice of actuator.
- Packet Tracer models a single global environment, so all sensors observe the same ambient conditions. A real facility would place independent sensors in several zones of the cold room and compare them, which also gives sensor failure detection by disagreement.
- The rule engine holds state only in the current sensor values. It has no time dimension, so it cannot express a rule such as "alarm if the temperature has been above 8 degrees for more than thirty minutes", which is how real cold chain standards are written. Implementing that would require a microcontroller board running Python or JavaScript inside the simulation.
- No SMS or email notification path exists, because Packet Tracer has no such service. The escalation is limited to the local siren and the two dashboards.
- WPA2-PSK with a shared passphrase across all nine devices is adequate for a teaching simulation but weak for production, where per device certificates under 802.1X would be used.

- The design has no redundancy. The IoT server is a single point of failure for the rule engine, and there is no modelled backup power for the cooling equipment.

9.4 Future scope

- Add a microcontroller board with custom code to implement time based excursion rules and to keep a rolling log of the temperature history for audit.
- Add a solar panel and battery to the model so that behaviour under mains failure, which is the single most common real cause of cold chain loss, can be simulated.
- Extend the design to several cold rooms across a district, each with its own access point and VLAN, reporting to one central server.
- Add an SMS or email gateway in the cloud tier so that alerts reach staff who are off site.
- Introduce redundant temperature sensors and a voting rule so that a failed sensor is detected rather than trusted.

10 Conclusion

This project designed, configured and successfully simulated SmartVax Cold Chain Guard, an IoT based monitoring and control system for vaccine cold storage, entirely within Cisco Packet Tracer. The implementation comprises nine IoT end devices, four sensing and five actuating, connected over a WPA2-PSK protected wireless access point into a switched and routed network of two subnets, with an IoT registration server providing DHCP, DNS, the device registry, the automation rule engine and the monitoring dashboard.

Twelve automation rules were implemented against a requirement of one, including a two sensor composite rule and a full set of paired reset rules. In free running simulation the system settled into a stable cycle between **6.0 and 8.3 °C**, holding the vaccine storage band without any operator action, ventilating automatically on high humidity, recording and illuminating every physical access to the stock, and raising an audible alarm on heat excursion, on freezing and on smoke. Remote monitoring across the routed link was verified from a supervisor workstation on a separate subnet.

The escalation path was also exercised. During the warm up from the raised ambient the room reached 17.7 °C and the excursion alarm activated while the air conditioner was already running, which is exactly the behaviour required when cooling cannot keep up. Once the loop settled into its steady cycle the alarm stopped firing, because the room no longer crosses the 10 °C threshold. Fire detection was confirmed separately, with a smoke level of 0.961665 driving the siren on. Two rules were not seen to fire. The ventilation off rule cannot, because the simulated ceiling fan has no effect on humidity. The composite door and temperature rule did not, because the tuned control loop leaves it only a 0.3 °C window to act in. Both are discussed in section 9.

The wider conclusion is that the change from periodic manual observation to continuous automated monitoring is not an incremental improvement. A paper chart filled in twice a day cannot detect an excursion that begins and ends between two readings, whereas this system detects and corrects it within one evaluation cycle. For a commodity as unforgiving and as consequential as a vaccine, that difference decides whether an entire batch reaches the patient in a usable state.

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Submitted files: `SmartVax_ColdChainGuard.pkt`, this report as PDF, and a screen recording of the simulation.